



(<http://www.pieriandata.com>)

Plotly and Cufflinks

Plotly is a library that allows you to create interactive plots that you can use in dashboards or websites (you can save them as html files or static images).

Installation

In order for this all to work, you'll need to install plotly and cufflinks to call plots directly off of a pandas dataframe. These libraries are not currently available through **conda** but are available through **pip**. Install the libraries at your command line/terminal using:

```
pip install plotly
pip install cufflinks
```

NOTE: Make sure you only have one installation of Python on your computer when you do this, otherwise the installation may not work.

Imports and Set-up

```
In [1]: import pandas as pd
import numpy as np
%matplotlib inline
```

```
In [3]: from plotly import __version__
        from plotly.offline import download_plotlyjs, init_notebook_mode, plot, iplot

        print(__version__) # requires version >= 1.9.0

3.10.0
```

```
In [4]: import cufflinks as cf
```

```
In [5]: # For Notebooks
        init_notebook_mode(connected=True)
```

```
In [6]: # For offline use
        cf.go_offline()
```

Fake Data

```
In [7]: df = pd.DataFrame(np.random.randn(100,4),columns='A B C D'.split())
```

```
In [8]: df.head()
```

Out[8]:

	A	B	C	D
0	0.372503	-0.609358	-1.671837	0.643126
1	-0.167786	0.370636	-0.510964	-0.740186
2	2.263638	-0.235863	-0.456551	-0.088189
3	0.688779	0.892185	-0.627377	1.609950
4	0.752993	-1.538034	0.695250	-0.268981

```
In [9]: df2 = pd.DataFrame({'Category':['A','B','C'],'Values':[32,43,50]})
```

```
In [10]: df2.head()
```

```
Out[10]:
```

	Category	Values
0	A	32
1	B	43
2	C	50

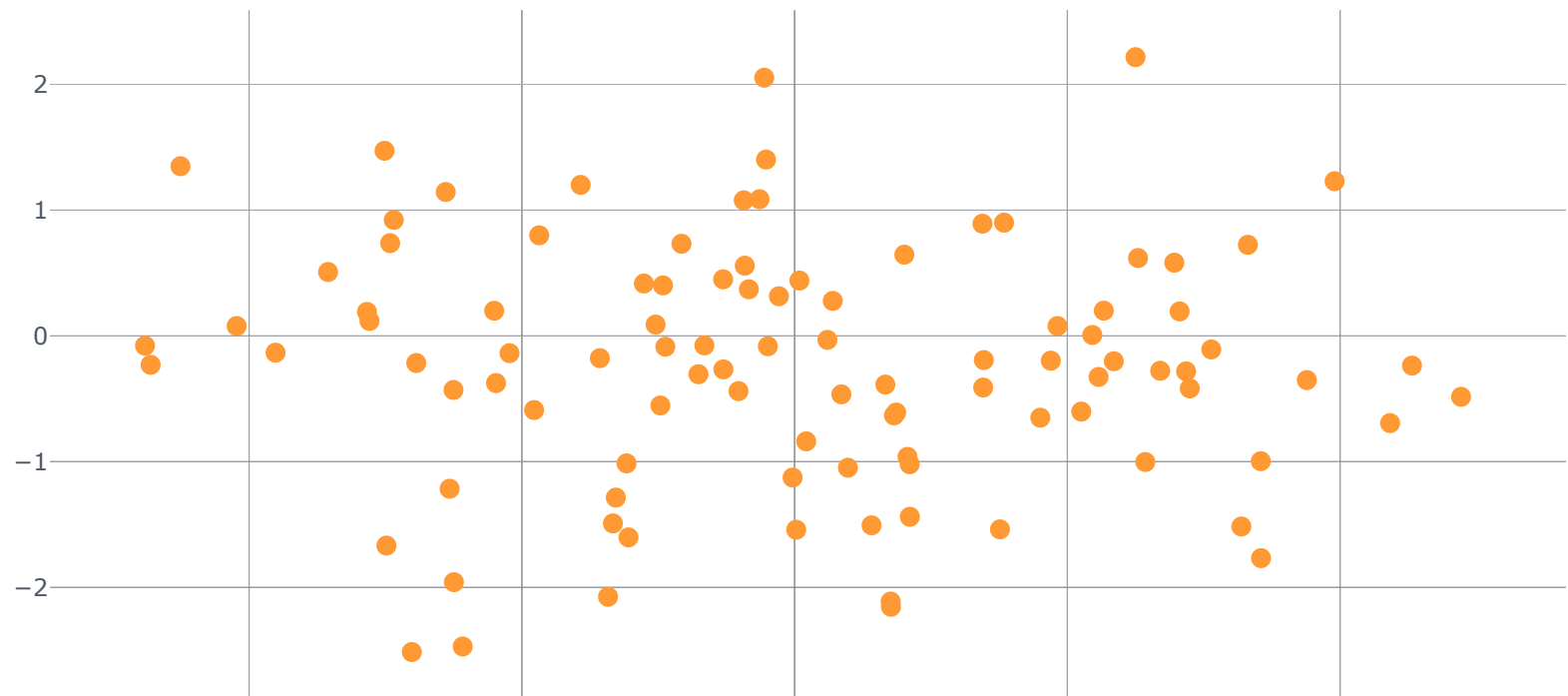
```
In [ ]:
```

Using Cufflinks and iplot()

- scatter
- bar
- box
- spread
- ratio
- heatmap
- surface
- histogram
- bubble

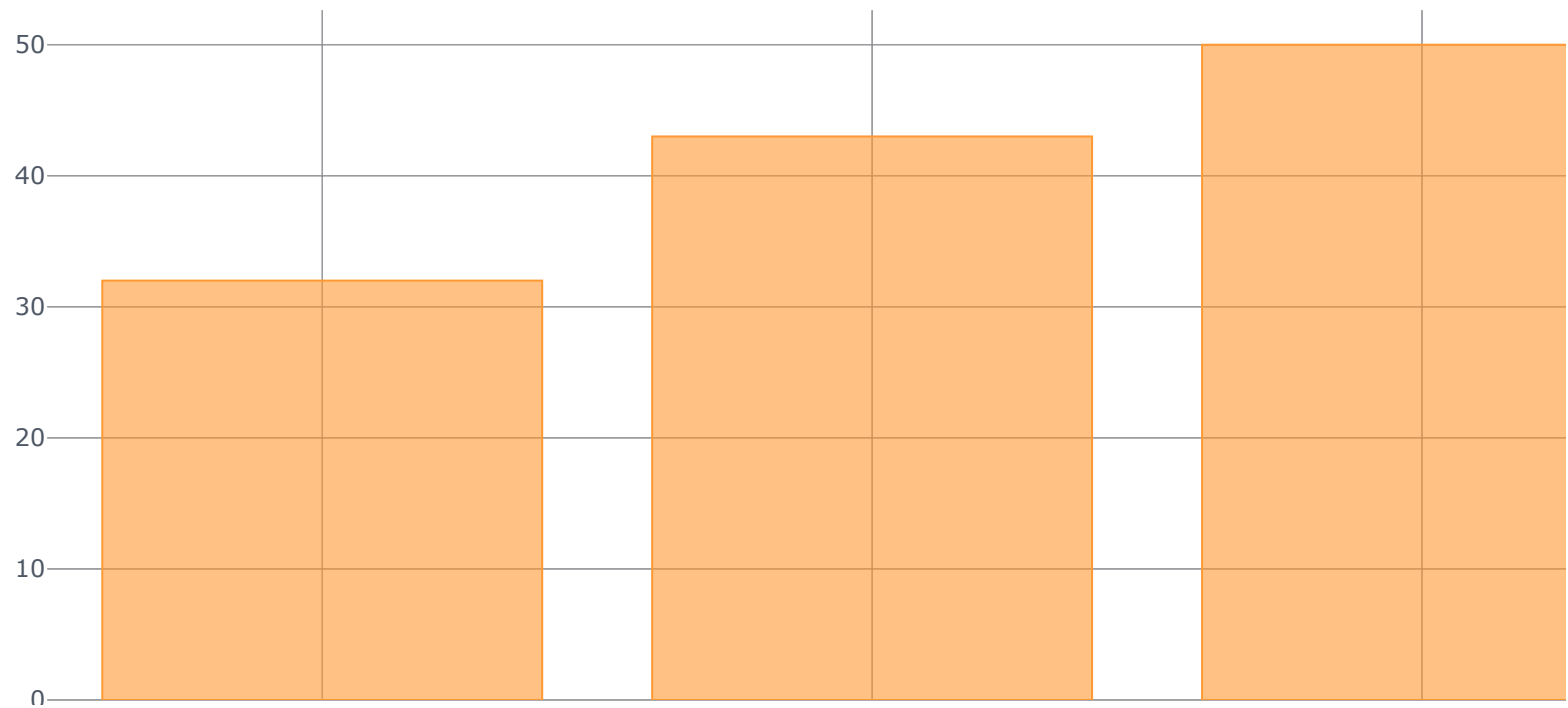
Scatter

```
In [11]: df.iplot(kind='scatter',x='A',y='B',mode='markers',size=10)
```

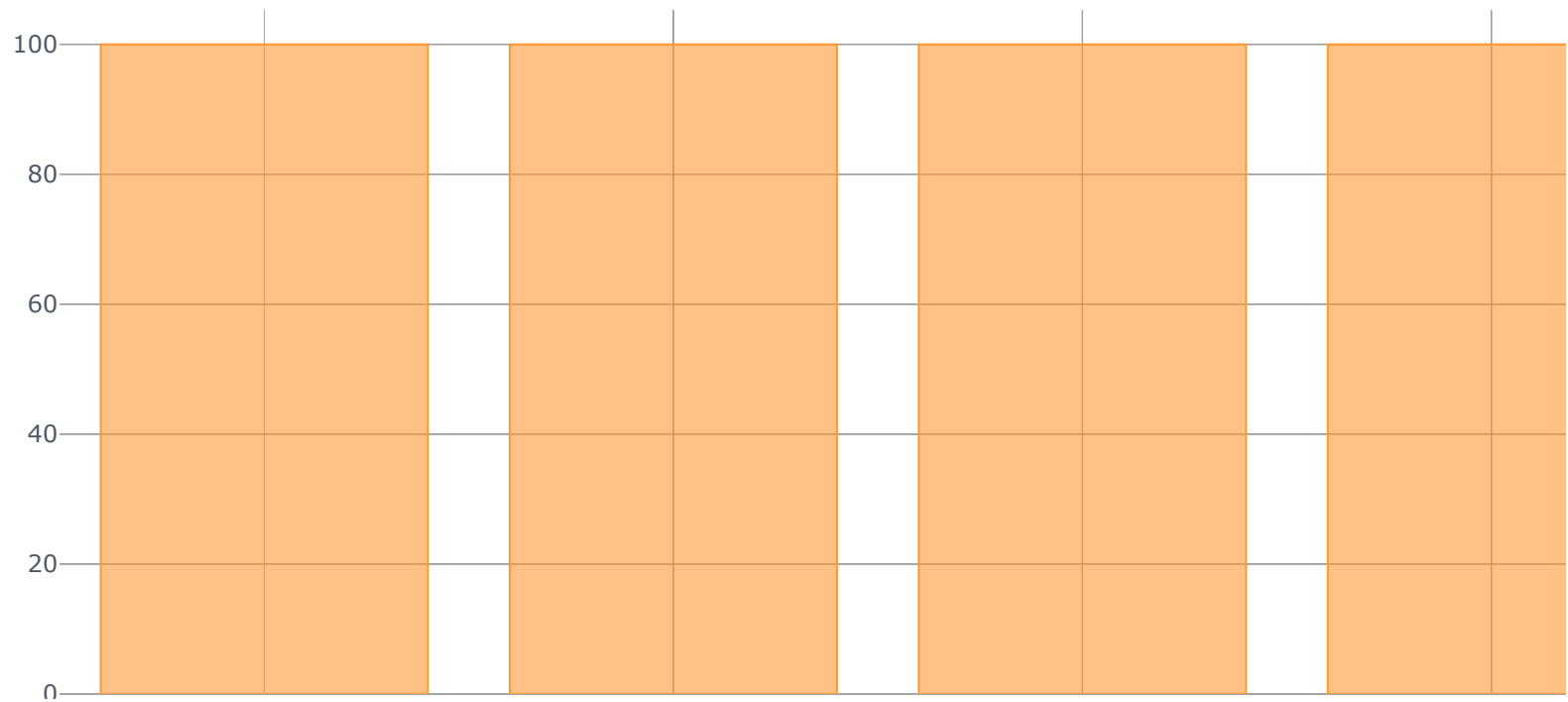


Bar Plots

```
In [12]: df2.iplot(kind='bar',x='Category',y='Values')
```

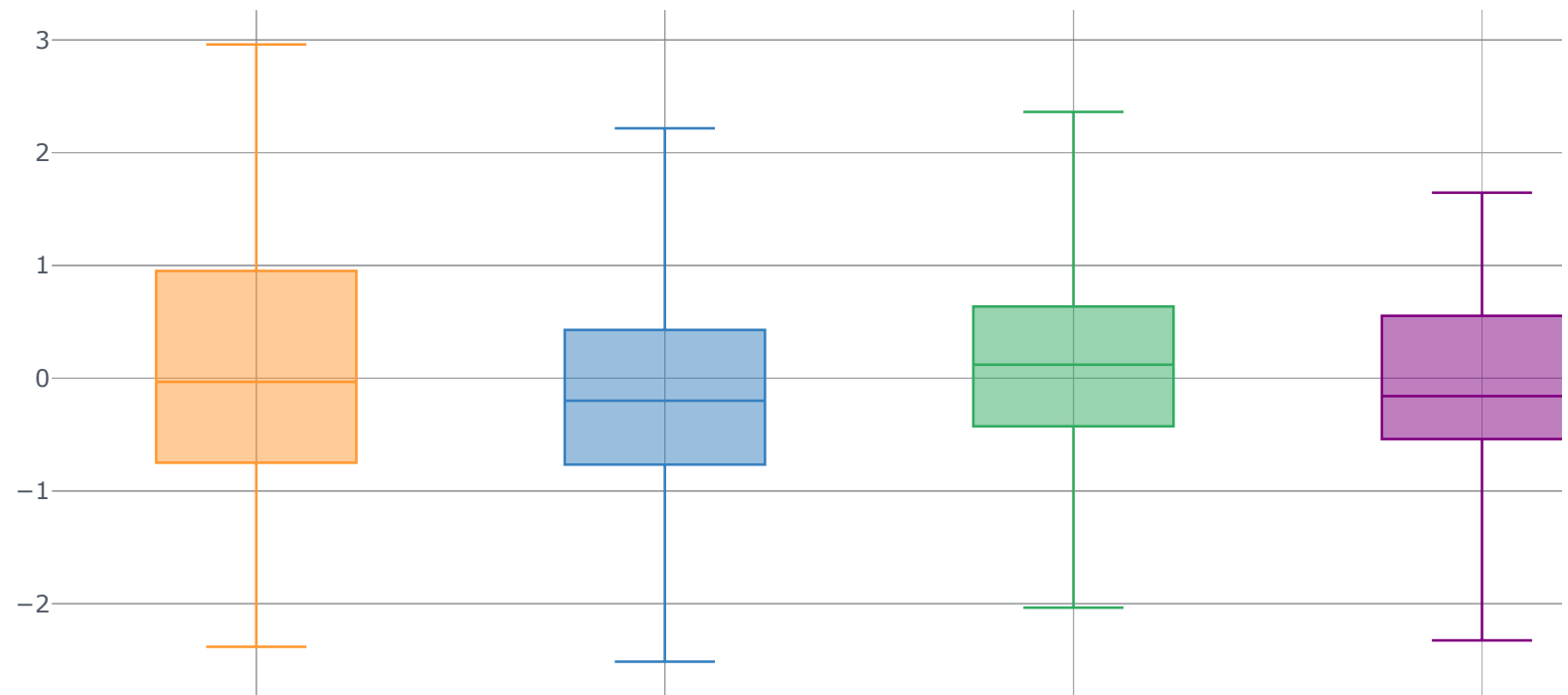


```
In [19]: df.count().iplot(kind='bar')
```



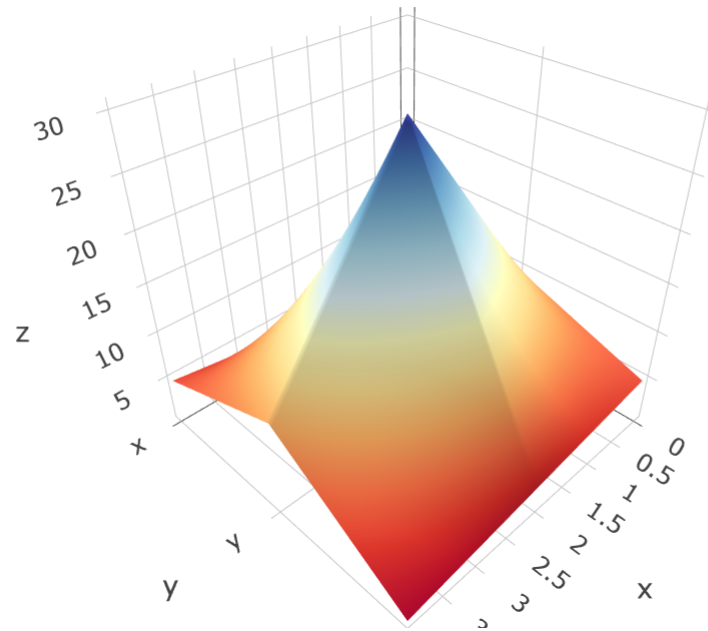
Boxplots

```
In [13]: df.iplot(kind='box')
```



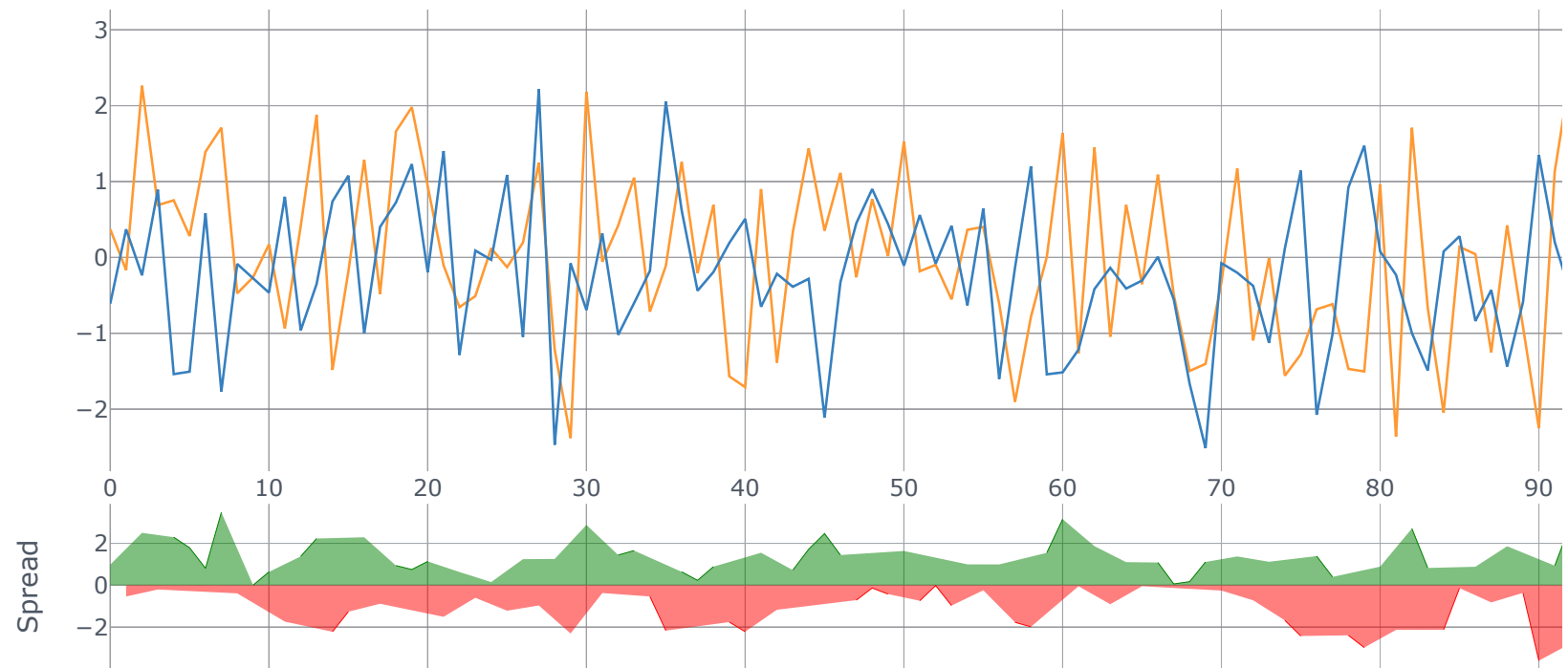
3d Surface

```
In [20]: df3 = pd.DataFrame({'x':[1,2,3,4,5], 'y':[10,20,30,20,10], 'z':[5,4,3,2,1]})  
df3.head()  
df3.iplot(kind='surface', colorscale='rdylbu')
```



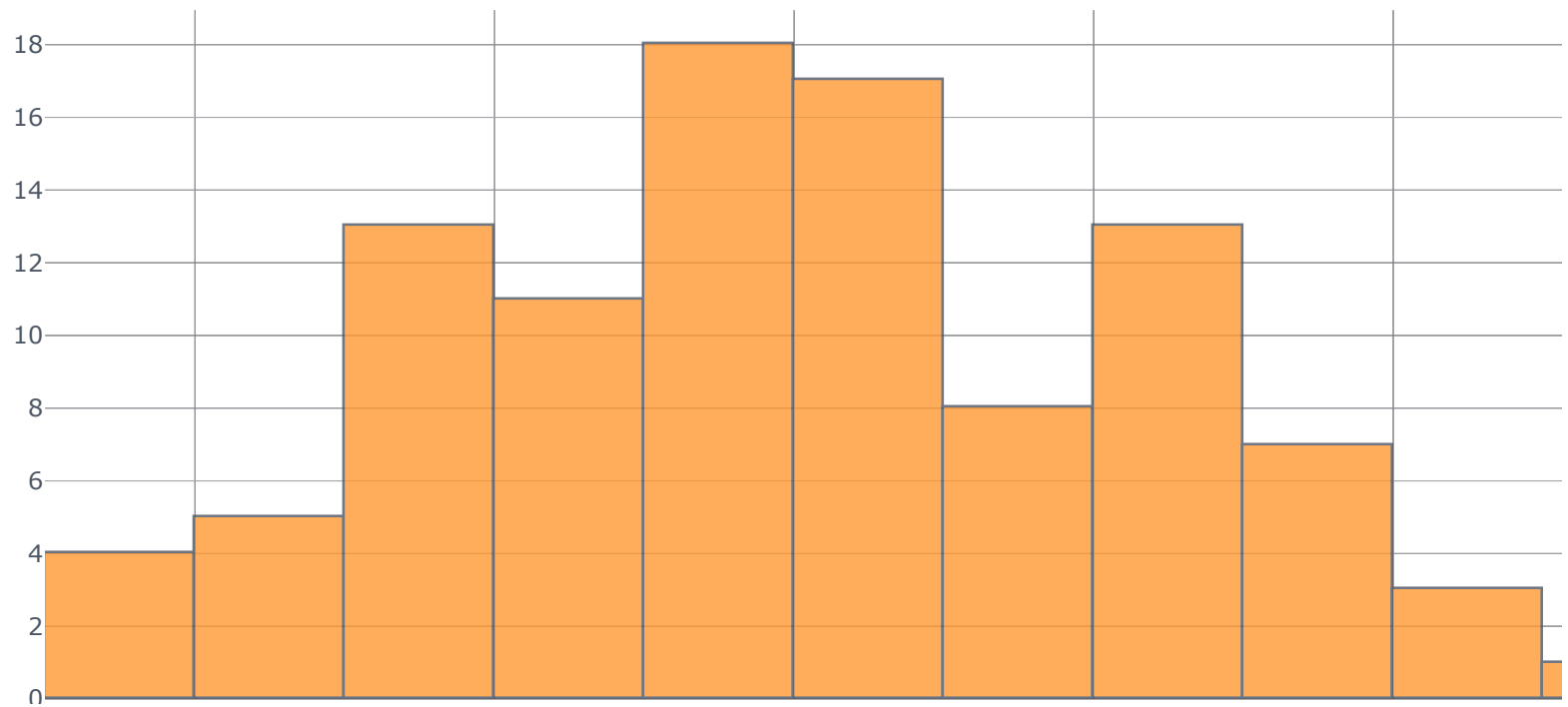
Spread

```
In [15]: df[['A', 'B']].iplot(kind='spread')
```

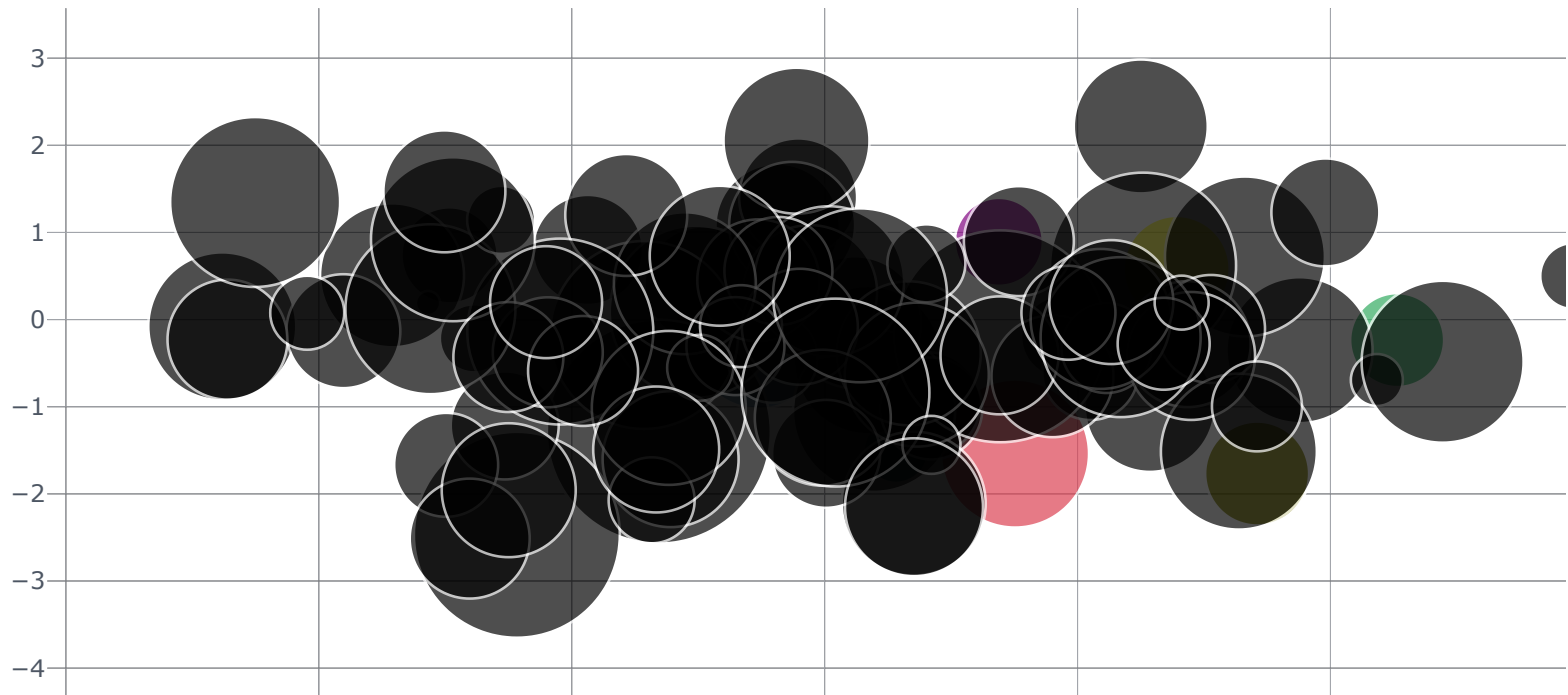


histogram

```
In [17]: df['A'].iplot(kind='hist',bins=25)
```



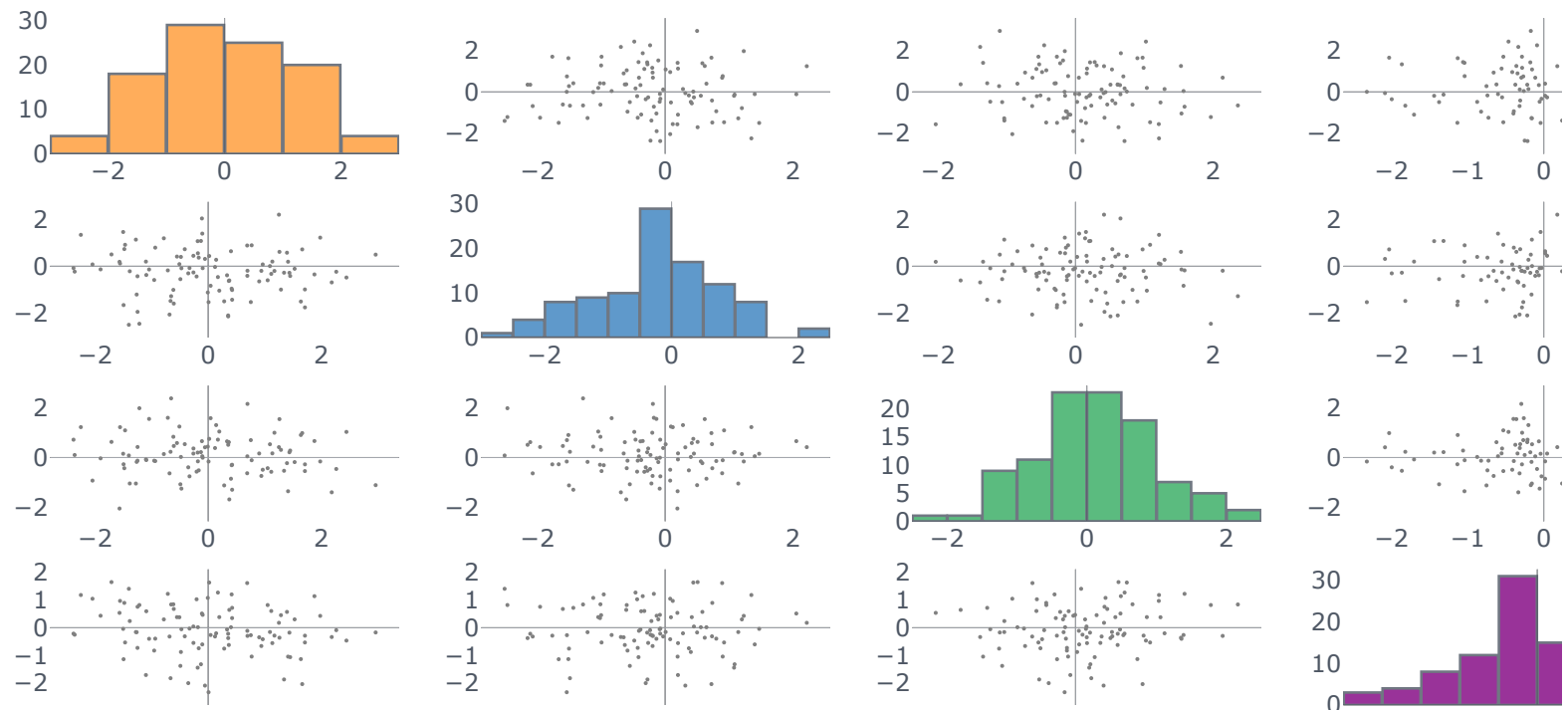
```
In [16]: df.iplot(kind='bubble',x='A',y='B',size='C')
```



scatter_matrix()

Similar to sns.pairplot()

```
In [18]: df.scatter_matrix()
```



Great Job!